

Investigating the filtering requirements of a single, bi-polar measurement channel

Background

Electrocardiography (ECG) is a method used to monitor the electrical activity of a patient's heart. ECGs are composed of three electrode placement systems. The focus of this lab was on the Standard Limb Lead placement¹, which is in the shape of an upside-down triangle (Einthoven's triangle). Each side of this triangle is called a Lead, and measures potential differences between the two points it connects. These two points are usually on two separate limbs, with one point carrying a positive electrode, and the other point (on the other limb) carrying the negative one. It is therefore said to be a "bi-polar" measurement. Since only one side of this triangle is being studied, it can also be said that it is a single measurement.

Before providing an electrocardiogram, ECG machines amplify and filter the electrical signals (recorded by the electrodes) of the heart's pumping activity². Filters are used to remove unwanted signals, and research shows that three filters are commonly used in ECG recordings: a high pass, a low pass, and a notch filter. High pass filters are used to block the DC offset (usually caused by electrode/skin/gel interface). Their cut-off frequency is around 0.05 Hz. Low pass filters are used to eliminate high frequency components which are not part of the ECG signal. These frequencies usually come from muscle tremors (natural contractions of the muscle, which cause movements inside the body). Their cut-off frequency is around 150 Hz. Finally, notch filters are used to remove mains (noise from the function generator for example, and distortion). Their cut-off frequency is at 50 Hz.

In a healthy patient, electrocardiograms traces can be divided into intervals, each representing the polarization or depolarization of different heart chambers. A first-degree heart block – or atrioventricular (AV) block – is when there is a delay in the time taken for the electrical pulse to move from the sinoatrial (SA) node in the upper heart chamber (atria), to the AV node in the lower part of the heart chamber (ventricle). On an ECG, it can be detected when the P-R interval is prolonged, therefore being equal to or over 200 milliseconds³.

Aim and Hypothesis

The aim of this experiment is to investigate the behavior, then test the effectiveness of the ECG's filtering system using pre-recorded data from several patients with suspected first-degree heart blocks.

As mentioned in the background, obtaining a cut off frequency of around 0.05 Hz for the high pass filter, 50 Hz for the notch filter, and 150 Hz for the low pass filter is expected. Behavior plots resembling the ones provided on Dr. Martin Fry's filter sheet were also anticipated:

<u>High pass filter</u>	<u>Low pass filter</u>	<u>Notch filter</u>
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¹ The Student Physiologist. Online Cardiology Resources. *The ECG Leads, Polarity, and Einthoven's Triangle*. Available at: <https://thephysiologist.org/study-materials/the-ecg-leads-polarity-and-einthovens-triangle/>

² Vetronic Services Ltd. Quality Anesthetic monitoring and Ventilation equipment. *Understanding the terminology used when ECG Monitoring*. Available at: <https://thephysiologist.org/study-materials/the-ecg-leads-polarity-and-einthovens-triangle/>

³ (2022). S. H. Oldroyd; B. S. Quintanilla Rodriguez; Amgad N. Makaryus. StatPearls. *First Degree Heart Block*. Available at: [https://www.ncbi.nlm.nih.gov/books/NBK448164/#:~:text=First%2Ddegree%20atrioventricular%20\(AV\),discovered%20only%20on%20routine%20ECG](https://www.ncbi.nlm.nih.gov/books/NBK448164/#:~:text=First%2Ddegree%20atrioventricular%20(AV),discovered%20only%20on%20routine%20ECG)

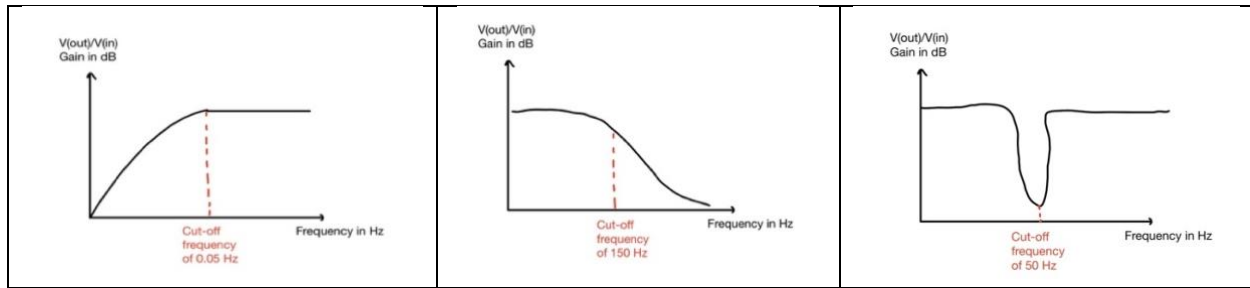


Table 1: Predicted filter behaviors

As for the effectiveness of the diagnosis test, obtaining a higher percentage of specificity compared to sensitivity can be predictable when studying a first-degree heart block (as this was the case in a previous study on using ECGs to detect right ventricular hypertrophy (RVH)⁴).

Method

The first part of this lab focused on characterizing the filters separately, using RLC circuit theory. To do so, finding the correct RC component values was necessary. A cut-off frequency can be calculated such that $f_c = \frac{1}{2\pi RC}$. By rearranging this equation, $RC = \frac{1}{2\pi f_c}$. Using the lab's available capacitor values, and the cut-off frequencies mentioned in the background for each filter, several resistor values were obtained. *View appendix for detailed calculations.*

High Pass Filter			Low Pass Filter			Notch Filter		
Capacitor	Resistor	RC	Capacitor	Resistor	RC	Capacitor	Resistor	RC
1.00E-09	3.18E+09	3.18E+00	1.00E-09	1.06E+06	1.06E-03	1.00E-09	3.18E+06	3.18E-03
2.20E-09	1.45E+09	3.18E+00	2.20E-09	4.82E+05	1.06E-03	2.20E-09	1.45E+06	3.18E-03
4.70E-09	6.77E+08	3.18E+00	4.70E-09	2.26E+05	1.06E-03	4.70E-09	6.77E+05	3.18E-03
1.00E-08	3.18E+08	3.18E+00	1.00E-08	1.06E+05	1.06E-03	1.00E-08	3.18E+05	3.18E-03
4.70E-08	6.77E+07	3.18E+00	4.70E-08	2.26E+04	1.06E-03	4.70E-08	6.77E+04	3.18E-03
1.00E-07	3.18E+07	3.18E+00	1.00E-07	1.06E+04	1.06E-03	1.00E-07	3.18E+04	3.18E-03
4.70E-07	6.77E+06	3.18E+00	4.70E-07	2.26E+03	1.06E-03	4.70E-07	6.77E+03	3.18E-03
1.00E-06	3.18E+06	3.18E+00	1.00E-06	1.06E+03	1.06E-03	1.00E-06	3.18E+03	3.18E-03
4.70E-06	6.77E+05	3.18E+00	4.70E-06	2.26E+02	1.06E-03	4.70E-06	6.77E+02	3.18E-03
1.00E-05	3.18E+05	3.18E+00	1.00E-05	1.06E+02	1.06E-03	1.00E-05	3.18E+02	3.18E-03
4.70E-05	6.77E+04	3.18E+00	4.70E-05	2.26E+01	1.06E-03	4.70E-05	6.77E+01	3.18E-03
1.00E-04	3.18E+04	3.18E+00	1.00E-04	1.06E+01	1.06E-03	1.00E-04	3.18E+01	3.18E-03
4.70E-04	6.77E+03	3.18E+00	4.70E-04	2.26E+00	1.06E-03	4.70E-04	6.77E+00	3.18E-03
1.00E-03	3.18E+03	3.18E+00	1.00E-03	1.06E+00	1.06E-03	1.00E-03	3.18E+00	3.18E-03

For a chosen capacitor value of $47\mu F$, the high pass filter's resistor value was $67.7 k\Omega$, the low pass's resistor value was 22.6Ω and the notch filter's resistor value was 67.7Ω . During the experiment, the values were rounded and adapted to the lab equipment provided ($68 k\Omega$ for the high pass, 22Ω for the low pass, and 68Ω for the notch).

After obtaining the required resistor and capacitor values, each filter was set up individually and voltage measurements (in and out) were taken. Then, the gain was obtained by computing $20 \times \log \left(\frac{V_{out}}{V_{in}} \right)$, and a figure representing the gain with respect to frequency was plotted. *All measurements are available in the appendix.*

Table 2 below shows the RC circuit diagrams of every filter, as well as their breadboard set up.

⁴ Prakash, R; 1981 Echocardiographic diagnosis of right ventricular hypertrophy: Correlation with ECG and necropsy findings in 248 patients, Catheterization and Cardiovascular Diagnosis 1981 7:2:179-184.

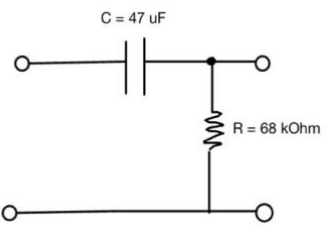
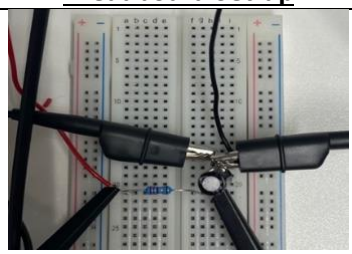
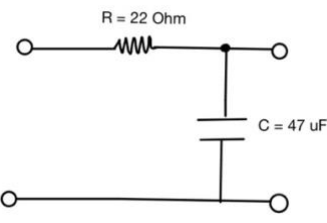
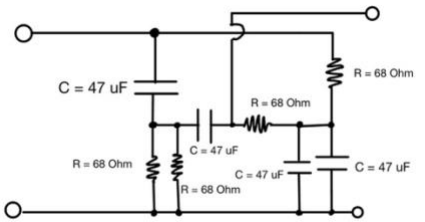
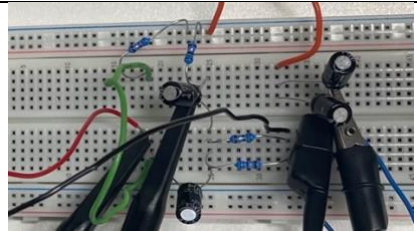
Filter	RC circuit diagram	Breadboard Set-up
High Pass		
Low Pass		
Notch		

Table 2: Circuit diagram and their corresponding breadboard set up

After completing this step for every filter, the three filters were combined into one large circuit, and the previous steps were repeated: voltage measurements were taken, gain calculations were made, and a final graph was plotted. Figure 1 below represents the full circuit's diagram (*its breadboard set up can be found in the appendix*).

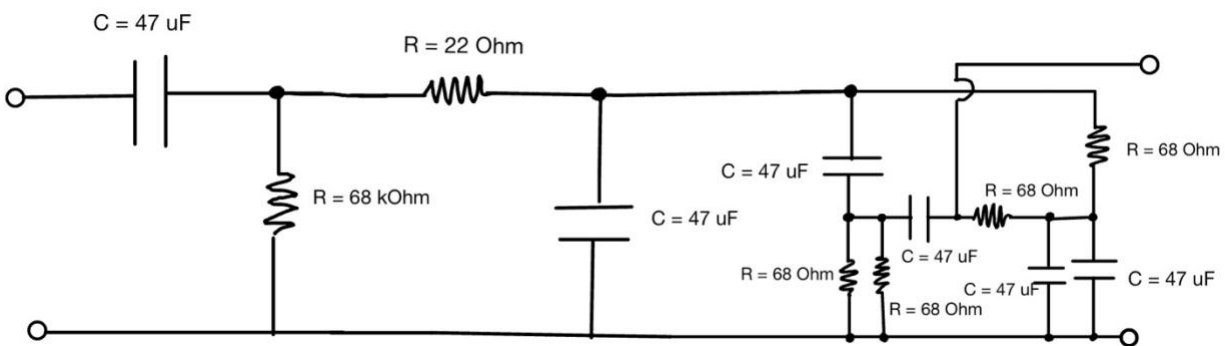


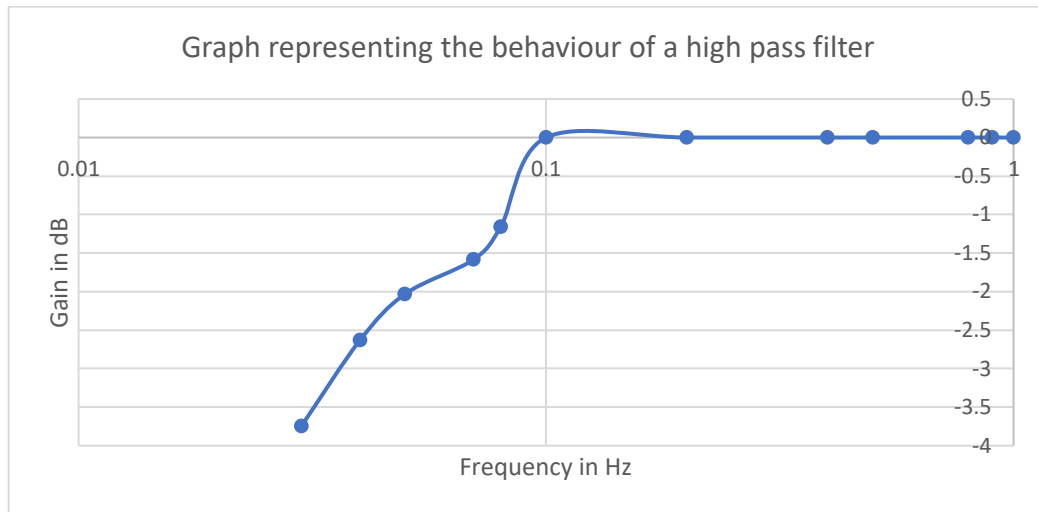
Figure 1: Filtering system with 3 filters combined

Following the full circuit set up, ECG traces were plotted using the provided MATLAB codes. *Command window input available in appendix*.

Lastly, the ECG traces were analyzed: first-degree heart block was determined in several patients, and the results were compared to the provided Gold Standard. The comparison results were then used to calculate the sensitivity and specificity percentages.

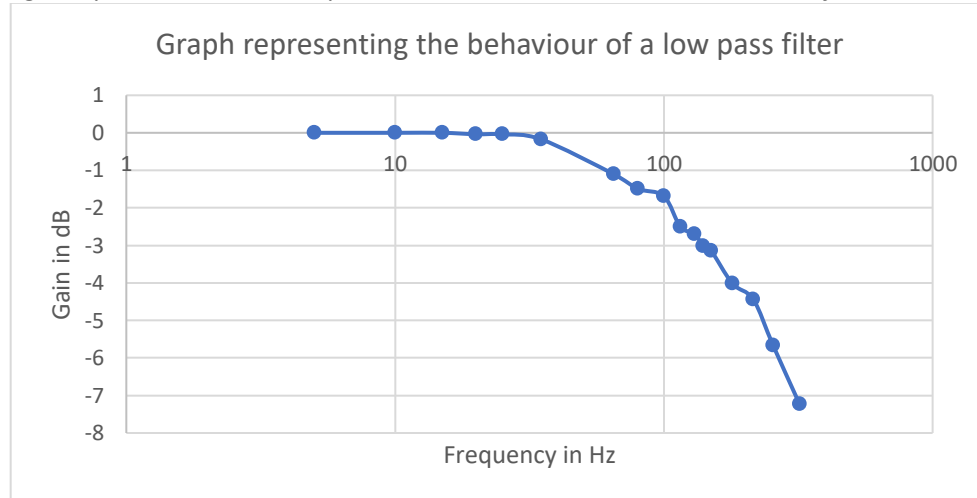
Results of Characterisation

The following high pass filter behavior plot was obtained. *Measurements can be found in the appendix.*



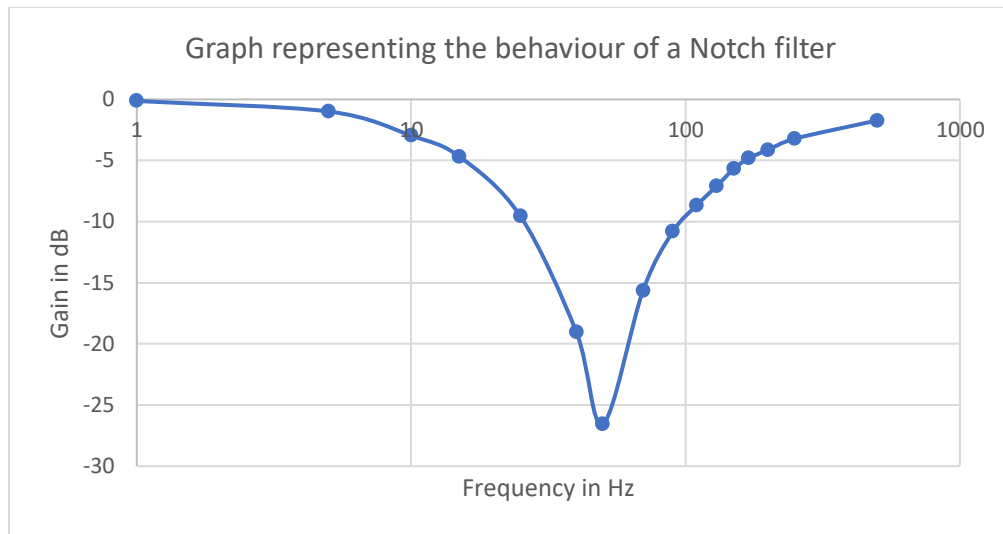
At -3 dB, the cut-off frequency was of around 0.035 Hz, which is close to the theoretical cut-off frequency. High pass filters function on very low frequency ranges, where the signal is easily disrupted by noise. Moreover, initial calculations showed that the resistor associated with the $47\mu F$ capacitor had a value of around 67.7 k Ω . Since a resistor of 68 k Ω was used instead, the slight value difference, along with the capacitor and resistor tolerances, could be another reason why the cut-off frequency was not accurate. Overall, however, the plot obtained confirms the hypothesis.

The following low pass filter behavior plot was obtained. *Measurements can be found in the appendix.*



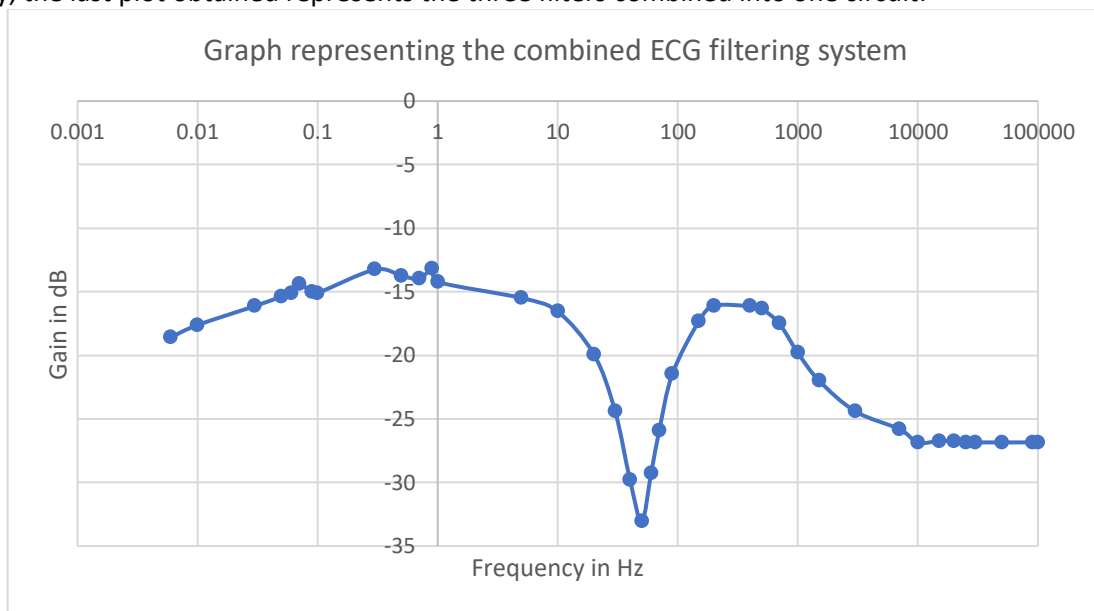
At -3 dB, the cut-off frequency was of around 150 Hz, which is equal to the theoretical cut-off frequency. Moreover, the plot is similar to Dr. Martin Fry's figure. The low pass filter hypothesis was therefore confirmed.

The following notch filter behavior plot was obtained. *All measurements can be found in the appendix*



The cut-off frequency was of 50 Hz, which is equal to the theoretical cut-off frequency. The plot also has the adequate shape, therefore confirming the hypothesis.

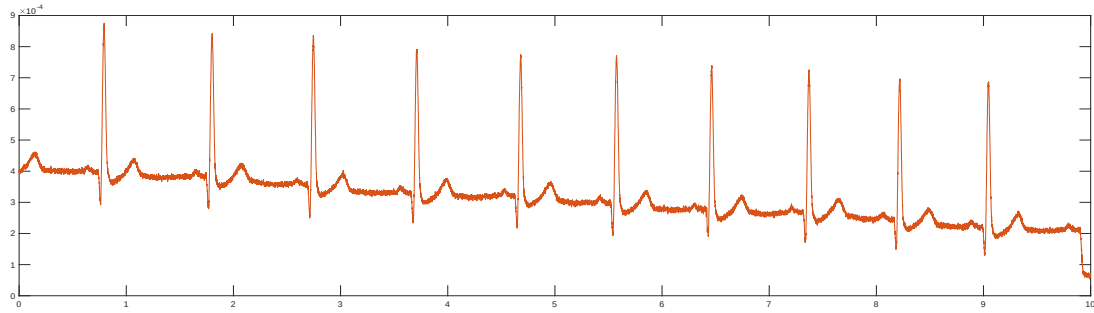
Finally, the last plot obtained represents the three filters combined into one circuit:



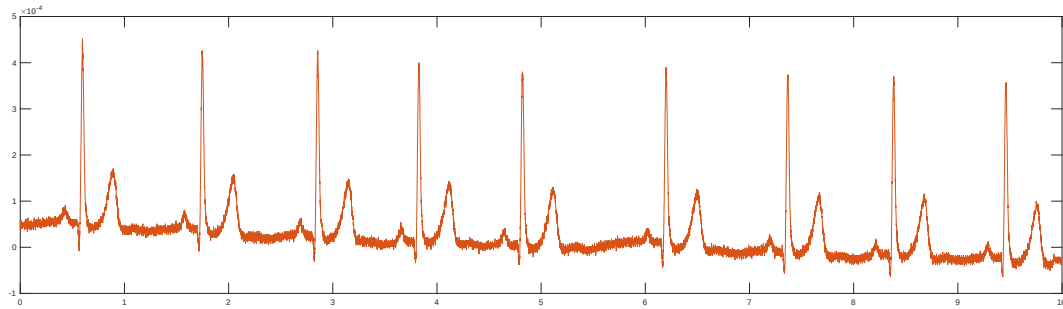
The pattern of every filter is still distinguishable. The notch filter still has a cut-off frequency of 50 Hz; however, the cut-off frequencies of the high pass and low pass filters have changed (from 0.035 Hz to invisible for high pass, and from 150 to 400 Hz for low pass). As mentioned in the ECG Assignment instructions sheet, cascading 3 filters is non-ideal, as it promotes loading effects. Thus, it can be assumed that the cut-off frequency shift phenomena visible on the plot is due to the loading effect.

Results of ECG results

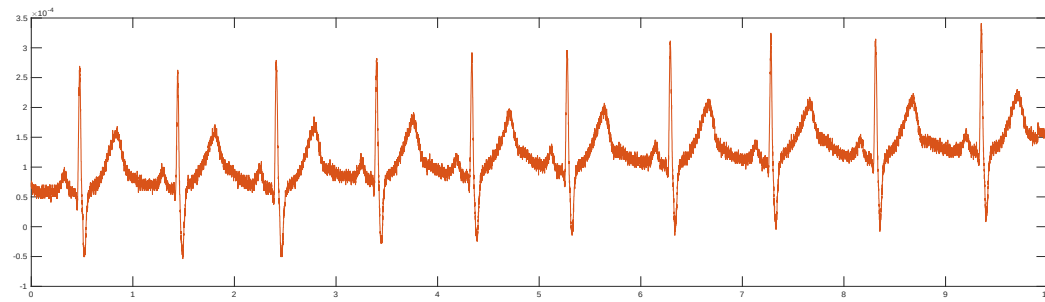
Once the combined filter circuit was set up, ECG recordings were taken. Below are three patient waveform examples:



Patient 13: PR Interval is normal. The patient is not sick.



Patient 20: PR interval is normal. The patient is not sick.



Patient 28: PR interval is ≥ 200 ms – First-degree heart block detected.

Three PR-intervals from three consecutive heartbeats were measured using MATLAB's Data Tips Tool, and the average value (in ms) was taken to determine whether the patient was sick or not. Below is a classification table of patients 1-30. The number 0 was used when the results were normal, and the number 1 was used when a first-degree heart block was detected.

Patient nb	Heartbeat 1 (in s)	Heartbeat 2 (in s)	Heartbeat 3 (in s)	Average PR (in ms)	Status	Patient nb	GOLD STANDARD	ECG Diagnosis	Test
1	0.3955-0.2335=0.162	1.2185-0.996=0.2225	1.9945-1.7975=0.197	193.33	normal	1	0	0	True negative
2	6.102-5.806=0.296	NO PR INTERVAL	7.763-7.5455=0.2175	256.75	PR≥200ms	2	0	1	False positive
3	0.549-0.43=0.119	1.417-1.2885=0.1285	2.261-2.1205=0.1405	129.33	normal	3	0	0	True negative
4	0.7135-0.4435=0.27	1.5635-1.2975=0.266	2.3865-2.084=0.302	279.33	PR≥200ms	4	0	1	False positive
5	1.922-1.6435=0.2785	3.0785-2.7835=0.295	4.194-3.963=0.231	268.17	PR≥200ms	5	1	1	True positive
6	0.6095-0.511=0.0985	1.391-1.287=0.104	2.152-2.049=0.103	101.83	normal	6	0	0	True negative
7	0.553-0.4335=0.1195	1.739-1.599=0.14	2.8855-2.773=0.1125	124	normal	7	0	0	True negative
8	1.5045-1.363=0.1415	2.8005-2.603=0.1975	3.8455-3.7025=0.143	160.67	normal	8	1	0	False negative
9	0.8235-0.722=0.1015	1.4835-1.382=0.1015	2.1235-2.0235=0.1	101	normal	9	0	0	True negative
10	5.0865-4.922=0.1645	5.9455-5.747=0.1985	6.7645-6.582=0.1825	181.83	normal	10	1	0	False negative
11	1.859-1.6765=0.1825	3.2165-3.023=0.1935	4.561-4.405=0.156	177.33	normal	11	0	0	True negative
12	1.123-0.9225=0.2005	2.003-1.8025=0.2005	2.881-2.663=0.218	206.33	PR≥200ms	12	1	1	True positive
13	0.7375-0.591=0.1465	1.7455-1.6035=0.142	2.692-2.537=0.155	147.83	normal	13	0	0	True negative
14	1.231-1.132=0.099	2.2915-2.1905=0.101	3.3515-3.267=0.0845	94.83	normal	14	0	0	True negative
15	1.1565-1.0165=0.14	1.7165-1.5765=0.14	2.3165-2.1965=0.12	133.33	normal	15	0	0	True negative
16	1.2475-1.099=0.1485	2.0875-1.9425=0.145	2.899-2.7225=0.1765	156.67	normal	16	1	0	False negative
17	0.926-0.7065=0.2195	1.753-1.5335=0.2195	2.5735-2.3635=0.21	216.33	PR≥200ms	17	1	1	True positive
18	5.635-5.4775=0.1575	6.5775-6.4215=0.156	7.45-7.25=0.2	171.16	normal	18	0	0	True negative
19	0.6085-0.475=0.1335	2.478-2.357=0.121	3.427-3.278=0.149	134.5	normal	19	0	0	True negative
20	0.53-0.343=0.187	1.6965-1.502=0.1945	2.8035-2.6205=0.183	188.17	normal	20	0	0	True negative
21	0.8155-0.694=0.1215	2.124-2.0025=0.1215	3.324-3.1935=0.1305	124.5	normal	21	0	0	True negative
22	0.5785-0.521=0.0575	1.159-1.0605=0.0985	1.715-1.619=0.096	84	normal	22	0	0	True negative
23	1.156-1.0355=0.1205	2.012-1.8915=0.1205	2.8445-2.749=0.0955	112.17	normal	23	0	0	True negative
24	1.178-1.0585=0.1195	1.689-1.6285=0.0605	2.2455-2.169=0.0765	85.5	normal	24	0	0	True negative
25	6.8315-6.6415=0.19	8.3685-8.1985=0.17	NO PR INTERVAL	180	normal	25	0	0	True negative
26	0.959-0.814=0.145	1.6615-1.512=0.1495	2.332-2.232=0.1	131.5	normal	26	0	0	True negative
27	1.4365-1.2665=0.17	2.409-2.2455=0.1635	3.277-3.1265=0.1505	161.33	normal	27	0	0	True negative
28	0.4275-0.215=0.2125	1.371-1.176=0.195	2.3665-2.132=0.2345	214	PR≥200ms	28	0	1	False positive
29	4.6795-4.5235=0.156	6.4195-6.243=0.1765	7.1415-6.9845=0.157	163.17	normal	29	0	0	True negative
30	6.6515-6.5135=0.138	7.253-7.1335=0.1195	7.849-7.7445=0.1045	120.67	normal	30	0	0	True negative

Classification of patients 1 to 30 on the left, and corresponding test results on the right.

Analysis

Diagnostic test	Definitive diagnosis		Total
	Abnormal (+)	Normal (-)	
Abnormal (+)	True positive = 3	False positive = 3	6
Normal (-)	False negative = 3	True negative = 21	24
Total	Total with conditions = 6	Total without conditions = 24	30

The sensitivity and specificity can be calculated:

- Sensitivity = True positive/Total with condition * 100 %
= 3/6 * 100 = **50%**
- Specificity = True negative/Total without condition * 100%
= 21/24 * 100 = **87.5%**

As anticipated in the hypotheses, a higher percentage of specificity than sensitivity was obtained. These percentages indicate that electrocardiography can discriminate patients who do have a first-degree heart block with patients who don't in a relatively accurate way (87.5%). However, a correct diagnosis can only be provided 50% of the time.

Discussion

1. Clinical and Ethical Implications

The percentages obtained for the sensitivity and specificity inform us that ECGs were able to show when a patient's PR interval was normal, however when the PR interval was prolonged, ECGs were ineffective in determining whether the disease was a FDHB or something else.

First-degree heart blocks can sometimes be confused with second-degree heart blocks, as both can be seen prolonged PR intervals on an ECG trace⁵. SDHBs are treated using pacemakers, which tend to cost around 4000 pounds. Their implanting procedures last around an hour, after which the patient could face several complications (infections) and limitations (with sports, or high electromagnetic field environments⁶).

Using an inaccurate test to determine what type of heart block the patient has can lead to the doctor suggesting a pacemaker where it is not needed, therefore costing the patient money and health complications unnecessarily.

This can be considered as unethical, as a healthy patient will find themselves getting treated for a disease they do not actually have. This could also cause them mental health and wellbeing problems.

For a sick patient, if the ECG test was not accurate and the FDHB was not detected in an early stage, the severity of the disease could increase and lead the to death.

Finally, on a clinical perspective, it would be inefficient for hospitals to invest in equipment that is imprecise. If several patients have been misdiagnosed, hospitals could get sued and face legal complications too.

2. Advantages and disadvantages of analog and digital Filters

In general, ECGs function with two types of filtering systems: analogue filters and digital filters.

The heart's electrical activity is first filtered and amplified using analog filters, which were the ones used during this lab. Advantages of analogue filters can be seen in the speed of the filtering, as well as the large frequency range the filters function in. Disadvantages of analog filters, however, are that they require a lot of space and are not easy to repair when facing technical problems (due to external factors, such as temperature or humidity for example).

After the analogue filtering is done, the signal is digitized to undergo digital filtering. Advantages of digital filtering is that they are programmable, therefore making them easy to test and build. They also offer more filtering precision and tend to be cheaper. Disadvantages, however, are that digital filters function slowly, and are unable to handle large frequencies very well.

Lastly, after being digitally filtered, the ECG trace can be seen on a screen or monitor.

Discussing the conversion process between analog and digital filtering is important. Since digital filtering strictly relies on the digitized signal obtained after the analogue filtering, if there were any inaccuracies made in the initial analog system, then they will be carried until the end of the data processing procedure, and flawed ECG traces will be obtained.

⁵ Healio Learn the Heart. *Second-degree atrioventricular (AV) Block Type 1 (Wenkebach) ECG Review*. Available at: <https://www.healio.com/cardiology/learn-the-heart/ecg-review/ecg-topic-reviews-and-criteria/second-degree-av-block-type-i-wenkebach-review#:~:text=Second%2Ddegree%20type%20%20AV,is%20an%20irregular%20R%2DR%20interval>.

⁶ Johns Hopkins Medicine. *Frequently Asked questions about pacemakers and Implantable Cardioverter Defibrillators*. Available at: <https://www.hopkinsmedicine.org/health/treatment-tests-and-therapies/frequently-asked-questions-about-pacemakers-and-implantable-cardioverter-defibrillators-icds#:~:text=When%20do%20I%20have%20to,overnight%20stay%20in%20the%20hospital>

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Appendix

Detailed calculations for the high pass filter. The same method is used for the low pass and the notch filter.

The cut off frequency of a high pass filter $f_c = 0.05 \text{ Hz}$. Since, $f_c = \frac{1}{2\pi RC} \Leftrightarrow RC = \frac{1}{2\pi f_c}$,

Then, $RC = \frac{1}{2\pi f_c} = \frac{1}{2\pi \cdot 0.05} \approx 3.18$.

Then, excel was used to input the different values of C in one column, and the formula $R = \frac{1}{2\pi f_c C} = \frac{1}{2\pi \cdot 0.05 C}$ in the second column. The third column computed RC to make sure 3.18 was always obtained.

By choosing $C = 47 \mu\text{F}$ and $R = 67.7 \text{ k}\Omega$, we can expect $f_c = 0.05002 \text{ Hz}$. However, since in reality $R = 68 \text{ k}\Omega$ was used, $f_c = 0.0498 \text{ Hz}$ was expected.

The same process was repeated for every filter.

Then, a constant capacitor value was chosen (any value available in the lab), as well as their corresponding resistor values.

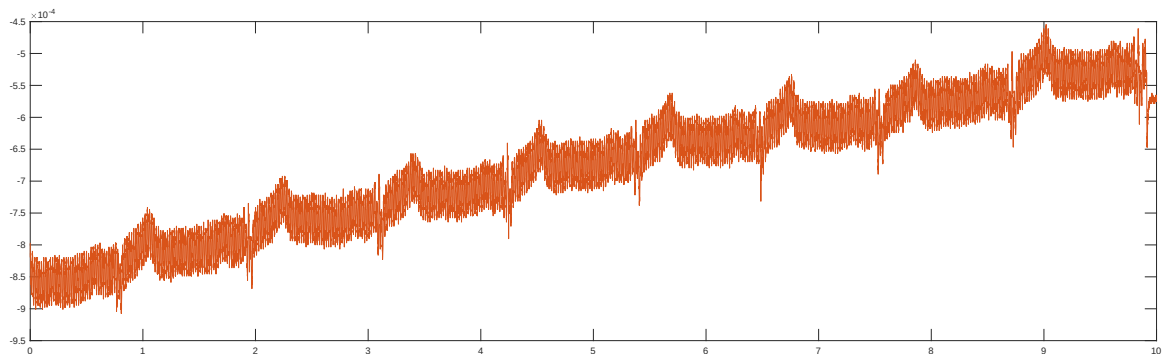
Measurements taken for the high pass, low pass, and notch filters.

High pass with R= 68kOhm and C= 47uF			
Vin	Vout	Frequency	Gain in dB
1.54	1.54	1	0
1.16	1.16	0.9	0
1.1	1.1	0.8	0
0.76	0.76	0.5	0
0.6	0.6	0.4	0
0.36	0.36	0.2	0
0.32	0.32	0.1	0
0.16	0.14	0.08	-1.159839
0.12	0.1	0.07	-1.583625
0.096	0.076	0.05	-2.029153
0.088	0.065	0.04	-2.631386
0.064	0.0416	0.03	-3.741733

Low pass with R=22 Ohms and C= 47uF			
Vin	Vout	Frequency	Gain in dB
4	4	5	0
4.56	4.56	10	0
4.56	4.56	15	0
4.48	4.46	20	-0.038863
4.36	4.34	25	-0.039935
4	3.92	35	-0.175478
3.36	2.96	65	-1.100951
3.04	2.56	80	-1.492672
2.72	2.24	100	-1.686418
2.56	1.92	115	-2.498775
2.4	1.76	130	-2.693971
2.32	1.64	140	-3.012883
2.24	1.56	150	-3.142468
2.16	1.36	180	-4.018297
2	1.2	215	-4.436975
1.92	1	255	-5.666025
1.84	0.8	320	-7.234557

Notch with R=68 ohms and C= 47 uF			
Vin	Vout	Frequency	Gain in dB
1.56	1.28	500	-1.718293
1.68	1.16	250	-3.217026
1.8	1.12	200	-4.12109
1.88	1.08	170	-4.814682
1.96	1.02	150	-5.673118
2.08	0.92	130	-7.08551
2.22	0.82	110	-8.650782
2.36	0.68	90	-10.80806
2.54	0.42	70	-15.63169
2.72	0.128	50	-26.54718
2.86	0.32	40	-19.02432
3.06	1.02	25	-9.542425
3.36	1.96	15	-4.681664
3.6	2.56	10	-2.961251
3.72	3.32	5	-0.988097
1.36	1.34	1	-0.128682

Example of inaccurate ECG traces, which show the system was not always perfect and reliable.

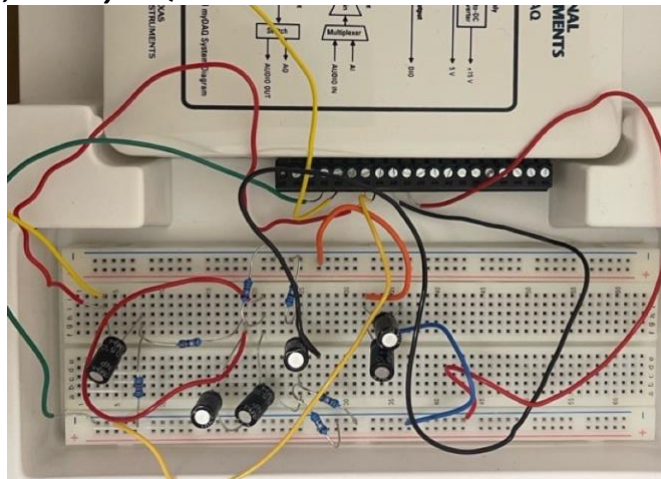


Patient 5: PR interval was difficult to analyze.

Measurements taken for the three filters combined.

All together			
Vin	Vout	Frequency	Gain in dB
1.76	0.08	100000	-26.848454
1.76	0.08	90000	-26.848454
1.76	0.08	50000	-26.848454
1.76	0.08	30000	-26.848454
1.76	0.08	25000	-26.848454
1.74	0.08	20000	-26.749185
1.74	0.08	15000	-26.749185
1.76	0.08	10000	-26.848454
1.56	0.08	7000	-25.800692
1.66	0.1	3000	-24.402162
1.76	0.14	1500	-21.987693
1.76	0.18	1000	-19.804803
1.8	0.24	700	-17.501225
1.84	0.28	500	-16.353196
2	0.312	400	-16.137508
2.4	0.376	200	-16.100468
2.64	0.36	150	-17.306029
3.32	0.28	90	-21.479601
3.64	0.184	70	-25.925671
3.72	0.128	60	-29.266659
3.96	0.088	50	-33.06425
3.96	0.128	40	-29.809704
4.24	0.256	30	-24.382518
4.52	0.456	20	-19.923472
4.52	0.672	10	-16.555383
4	0.672	5	-15.493814
1.44	0.28	1	-14.224089
1.28	0.28	0.9	-13.201039
1	0.2	0.7	-13.9794
0.0896	0.0184	0.5	-13.749804
0.0736	0.016	0.3	-13.255157
0.182	0.032	0.1	-15.098428
0.158	0.028	0.09	-15.029981
0.126	0.024	0.07	-14.403186
0.114	0.02	0.06	-15.117497
0.094	0.016	0.05	-15.380157
0.064	0.01	0.03	-16.123599
0.038	0.005	0.01	-17.616272
0.0424	0.005	0.006	-18.567917

Final breadboard set up, with myDAQ connections.



MATLAB Command Window Input (example of patient 1).

```
Command Window
fx >> plot(time1, data1)
```